

Unit 3: Periodicity Worksheet:

1. **State** the missing element names and symbols:

Symbol	Element
H	
He	
Na	
O	
F	
Na	
Mg	
Al	
	Sulphur
	Chlorine
	Potassium
	Calcium
	Iron
	Copper
	Zinc
	Carbon

1. **State** the names of the following groups in the periodic table:

Group 1	
Group 2	
Group 17	
Group 18	

2. **Outline** what all elements in one group have in common?

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3. **Outline** what all the elements in one period have in common?

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4. **Outline** what happens to metal atoms to form cations? Give 3 examples, stating the electron configuration of the atoms and cations and state the charge on the cation.

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5. **Outline** what happens to non-metal atoms to form anions? Give 3 examples, stating the electron configuration of the atoms and anions and state the charge on the anion.

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6. **Outline** the trend in metallic behaviour across the periodic table, from left to right.

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7. **State** 3 metalloids.....

8. **State** 4 physical properties of metals

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9. **Explain** why alkali metals react strongly with halogens?

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10. **State and explain** the following trends that you have discovered in the periodic table:

a. Metal reactivity going down groups 1:

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b. Non metal reactivity going down group 17:

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c. Atomic radii across the groups:

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d. Atomic radii going down the periods:

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e. First ionization energy across the groups:

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f. First ionization energy going down the periods:

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g. Electronegativity across the groups:

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h. Electronegativity going down the periods:

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i. Which of the trends above are linked in your opinion? Explain your thinking.

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